Rationalisation

Read the game, then play the odds

In pairs
Two games, six rounds each. Read them.

→

Against the die
Three mixes, six rounds each.

→

Compare
Which reply was best, and why.

In pairs, one of you plays the rows; against the die, you play the **column**.

$$A = \begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix} \qquad B = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}$$

A pays the row player, B pays the column player. Rows are r_1 and r_2; columns are c_1 and c_2.

Every cell adds to ____, so this is a _____ game.

If the row player picks r_1 and you pick c_2, you score ____ and they score ____.

Decide who is the row player. Both choose at the same time, six times. Fill in every column.

The game: $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ $B = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$ rows r_1, r_2; columns c_1, c_2; A pays the row player, B the column player.

Round	a_1	a_2	u_1	u_2	$\sum u_1$	$\sum u_2$
1						
2						
3						
4						
5						
6						

Could either of you have done better by always playing the same action?

Same pairs, same six rounds. The payoffs are no longer mirror images, so read each cell carefully.

The game: $A = \begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix}$ $B = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}$ rows r_1, r_2; columns c_1, c_2; A pays the row player, B the column player.

Round	a_1	a_2	u_1	u_2	$\sum u_1$	$\sum u_2$
1						
2						
3						
4						
5						
6						

Did either of you settle on one action? Did it keep working?

The row player is a die now. Commit to your c_1 or c_2 *before* each roll. One D12 covers all three mixes.

The game: $A = \begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix}$ $B = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}$ rows r_1, r_2; columns c_1, c_2; A pays the row player, B the column player.

	$\sigma_1 = (1/4,\ 3/4)$ r_1 on 1 to 3			$\sigma_1 = (3/4,\ 1/4)$ r_1 on 1 to 9			$\sigma_1 = (1/3,\ 2/3)$ r_1 on 1 to 4		
Round	a_2	**die**	$\sum u_2$	a_2	**die**	$\sum u_2$	a_2	**die**	$\sum u_2$
1									
2									
3									
4									
5									
6									

Totals after six rounds:

Against $\sigma_1 = (p,\ 1-p)$ your two actions pay

$u_2(c_1, \sigma_1) =$ $u_2(c_2, \sigma_1) =$

So c_1 is the better reply exactly when p , and the two are equal at $p =$.

Is c_1 dominated by c_2?

Is c_2 dominated by c_1?

Neither, because which one is better depends on _____.

Dominance only settles an action that is worse against *every* strategy of the opponent. Here each action is a best response to some mixes, so we need the _____ instead.

Against the third die both of your actions pay _____, so every strategy of yours is a best response and your six-round total should be about _____.

That is the row player's half of the equilibrium: they roll precisely so that you cannot exploit them. Which of your own mixes does the same to *them*?

Dominance and best responses are not examined on their own. They are examined together with **Nash Equilibrium**, so the natural follow-up is **Question 1** on that page, where the best response condition does exactly this job for Rock-Paper-Scissors.

The habit worth building now: before computing anything, ask which actions could ever be a best response. Anything that never can be is not worth carrying into the algebra.

vknight.org/gt/topics/nash-equilibrium.html